MK vision
Smart Secure Style





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Smart Secure Style

Introduction

MK Vision was founded on **July 9, 2020.**

We provide **ODM/OEM** and **turnkey factory solutions**, driven by a team of Vietnam-based experts.

From **concept to mass production**, we turn your ideas into **scalable, high-quality** products with comprehensive support every step of the way.



MK Vision - Subsidiary of MK Group

MK Group is one of Vietnam's leading corporations in the field of security technology and identification.

As a member of MK Group, our mission is to become a pioneer in the smart security equipment industry in Vietnam and the region.

About MK Group

Over 25 years of growth, MK Group and its members have constantly optimized their strengths in mastering core technologies and developing main production and business pillars, including:

- **Smart Cards:** Manufacturing smart card products and developing smart card applications and eco-systems.

- **Secure ID Documents:** Manufacturing and providing a complete issuance solution for e-ID cards, e-passports, driver licenses, and other specialized documents.

- **Digital Security:** Supplying secure authentication solutions for government, banks and enterprises, etc.

- **AI Cameras:** manufacturing and supplying Artificial Intelligence (AI) camera systems for security surveillance, traffic management, fleet management, law enforcement, border control, etc.



Businesses

Built by MK Vision, Tailored for You



Turnkey Smart Factory Installation Services



Security Camera ODM/OEM

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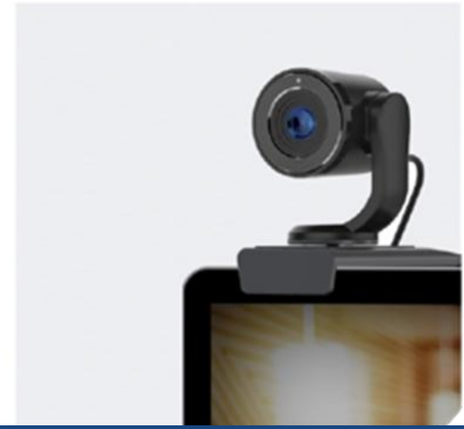




Make in Vietnam





Optimizing the value chain

Built by MK Vision, Tailored Everything you need to build an excellent product

IDEA

Products and Ideas

Market research

DESIGN

Mockup

Development

Prototype

SUPPLY CHAIN

Quality

Cost Optimize

MANU-FATURE

Mass product

Testings

SALE

Brand

Distribution

AFTER SALE

Customer service

Warranty period



Features & Technologies



Quality & Cost



Prices & Market



Meet our R&D team

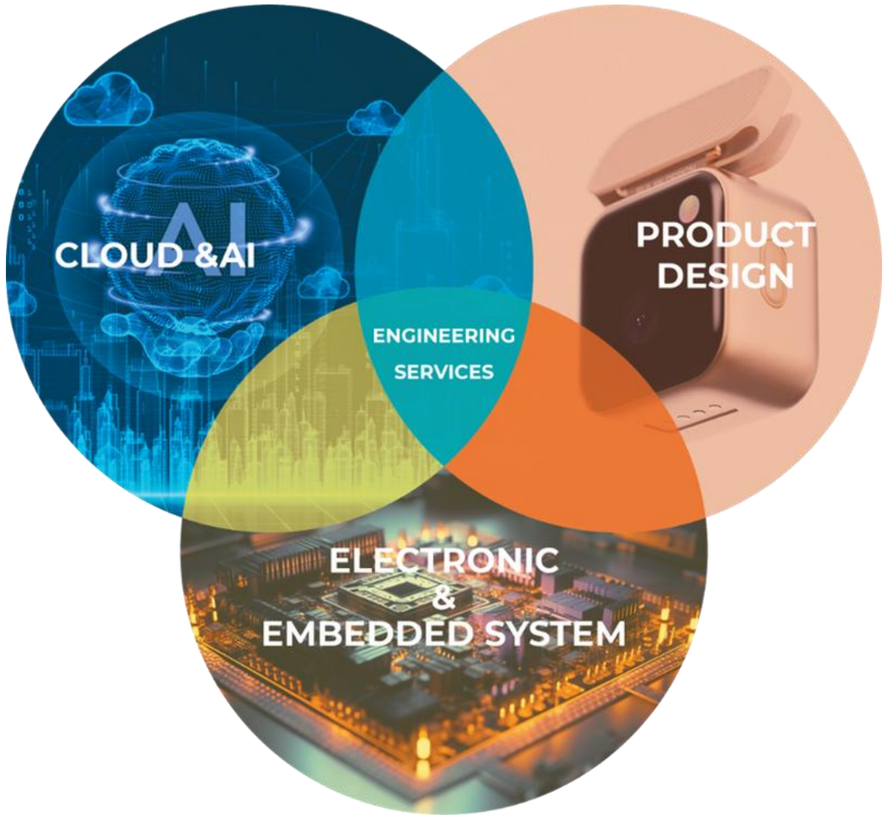
Innovation that make differences

- AI developer
- Electrical Engineer
- Project Manager
- Optical Engineer
- FW Engineer
- Mechanical Engineer
- Cloud & Platform Developer
- RF Engineer

Our end-to-end ODM services

Cloud & AI

Mobile & desktop apps
UX/UI & Web
Cloud services



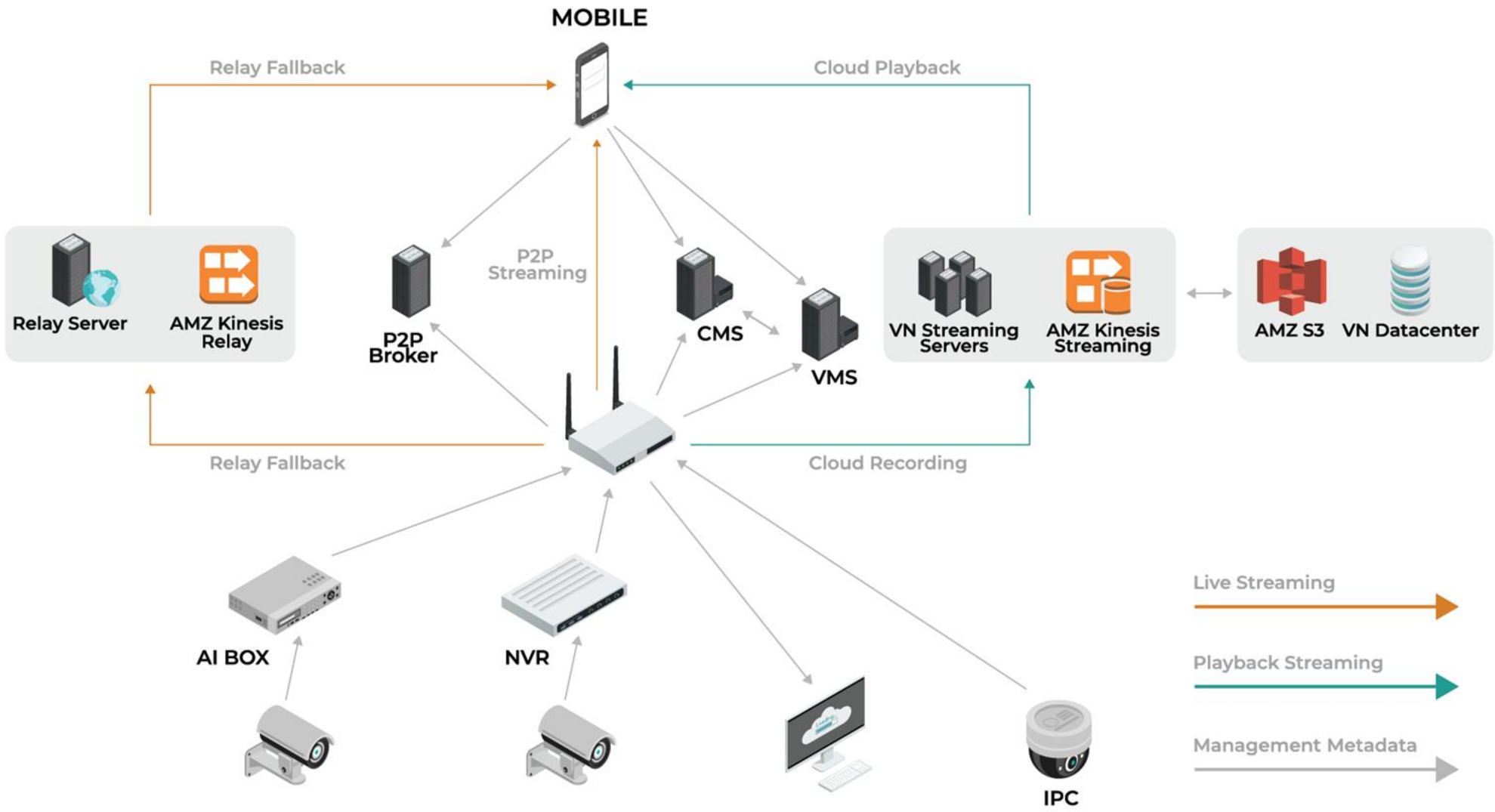
Product Design

Industrial design
Structure design & simulation
Tooling

Electronic & Embedded System

PCBA
Firmware
Valuation test & NPI

Smart CCTV System Platform



Industrial Design

We don’t just design. We create experiences

Our industrial design team brings strong expertise in developing **smart home devices**, **security cameras**, and **a wide range of electronic products**.



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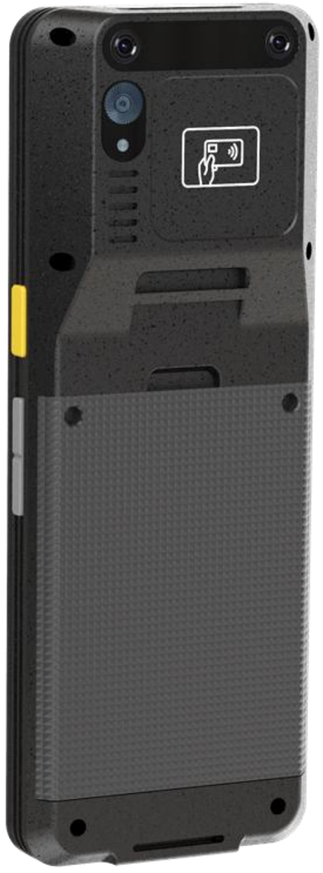


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Industrial Design

We don’t just design. We create experiences

From product design to packaging design, we ensure every detail aligns with your **brand vision**, reflects your **quality**, and enhances **brand awareness**.



Hardware Capabilities

We make it work — Fast, Strong, Reliable



Platform

- Ambarella CV25, Cv72
- Ingenic T31

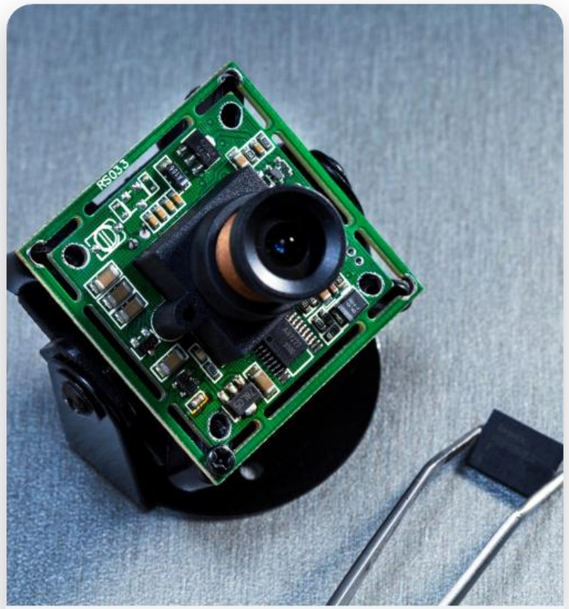


Image Sensor

Sony sensor IMX327, IMX335, IMX415, IMX 485, IMX264 global shutter

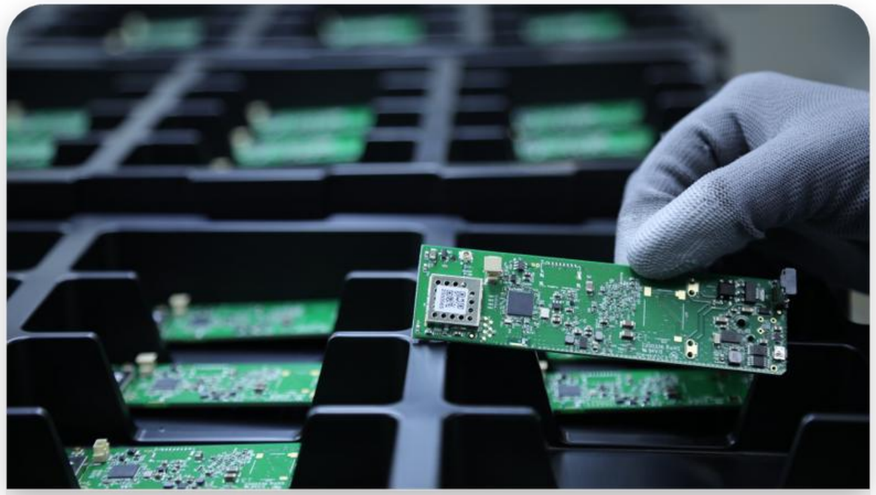


Multi-Sensor support

Support 4 image sensor

Hardware Capabilities

We make it work — Fast, Strong, Reliable



Design Services

•Schematic & PCB: Altium Designer
•Simulation: Signal Integrity SI/PI tools
•3D Enclosure Integration: SolidWorks, STEP export
•Manufacturing Output: Gerber, BOM, Pick & Place, Assembly drawing, Test points map



DFM, DFT

Design for Manufacturability (DFM) and Design for Testability (DFT)

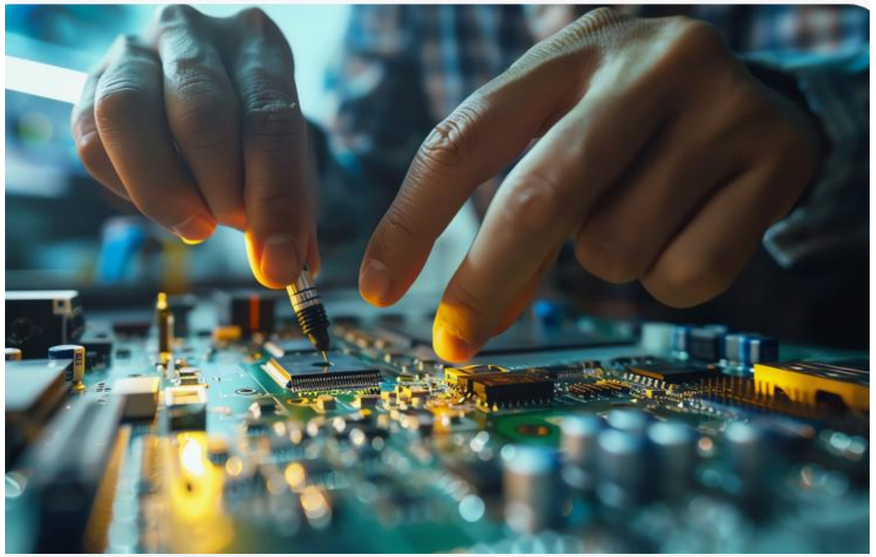
Hardware Capabilities

We make it work — Fast, Strong, Reliable



Testing & Validation Options

- Prototype bring-up and hardware validation
- Signal measurement with oscilloscope, logic analyzer, multimeter
- Environmental & stress testing
- Functional testing (FCT) and In-Circuit Test (ICT) jig development
- Firmware loading support



ICT, FCT

In-Circuit Test (ICT) & Functional Circuit Test (FCT) Services

Chipset Platforms

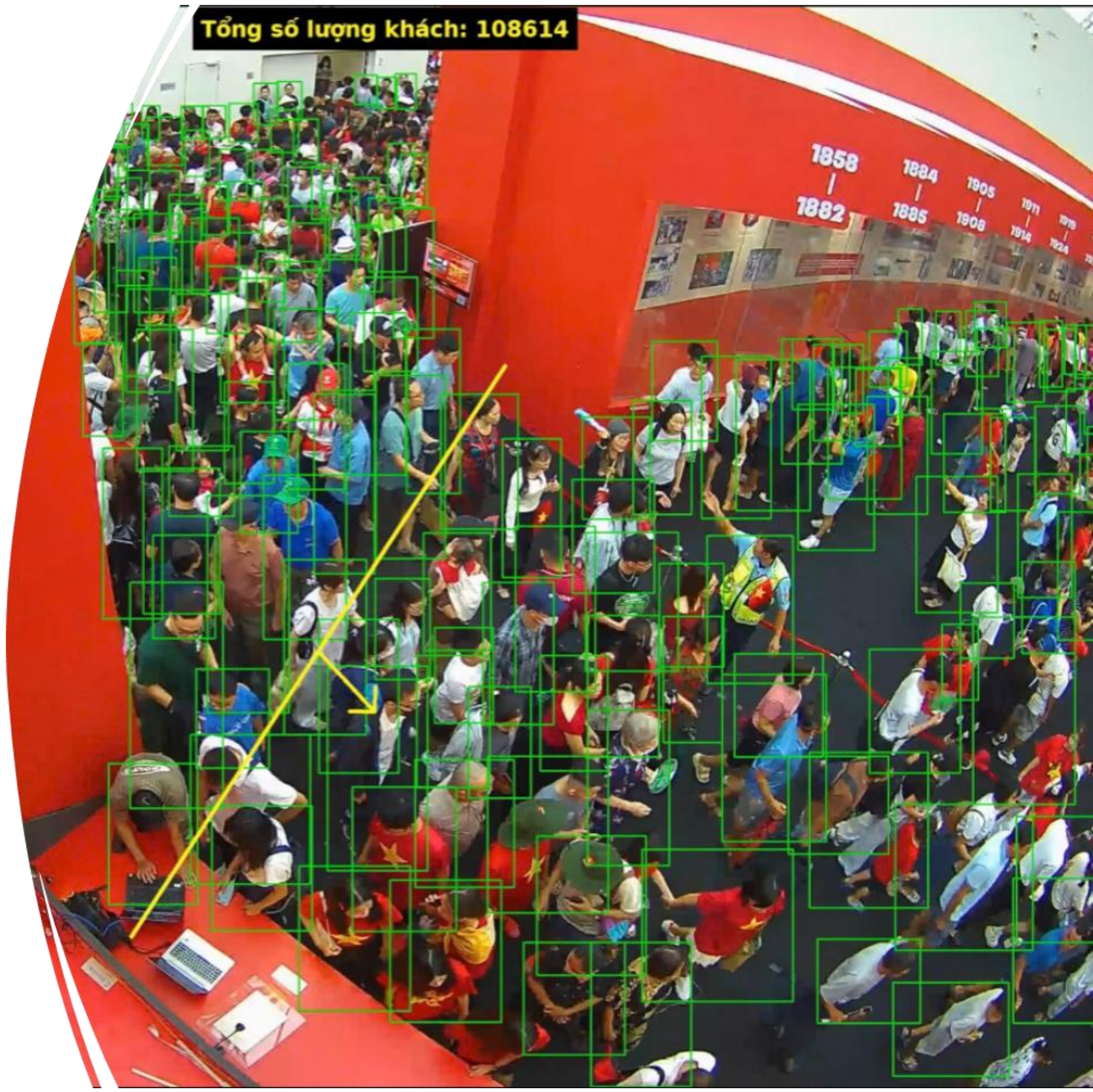


MK vision	Ambarella	Ingenic	Novatek
Camera OS	X	X	X
Camera SDK	X	X	
ONVIF	X		X
RTMP	X	X	
RTSP	X	X	X
HLS	X		
MQTT	X	X	
Qmulux Cloud		X	
WebRTC	X		
Built-in WebGUI	X		
Built-in network API	X	X	X
Human tracking	X	X	
Face Recognition	X		
Face Liveness	X		
Vehicle Recognition	X		
OCR	X		
Intrusion	X		
People/ Vehicle Counting	X		
MP4 Fragment	X	X	

Intelligent Firmware

Optimized for performance & AI

- **Multi-sensor & image/video optimization across** diverse hardware.
- **High security:** data encryption, secure boot.
- **Powerful Edge AI:** on-device recognition, classification, and analytics.
- **Modular SDK:** easy integration and custom application development.



AI capabilities

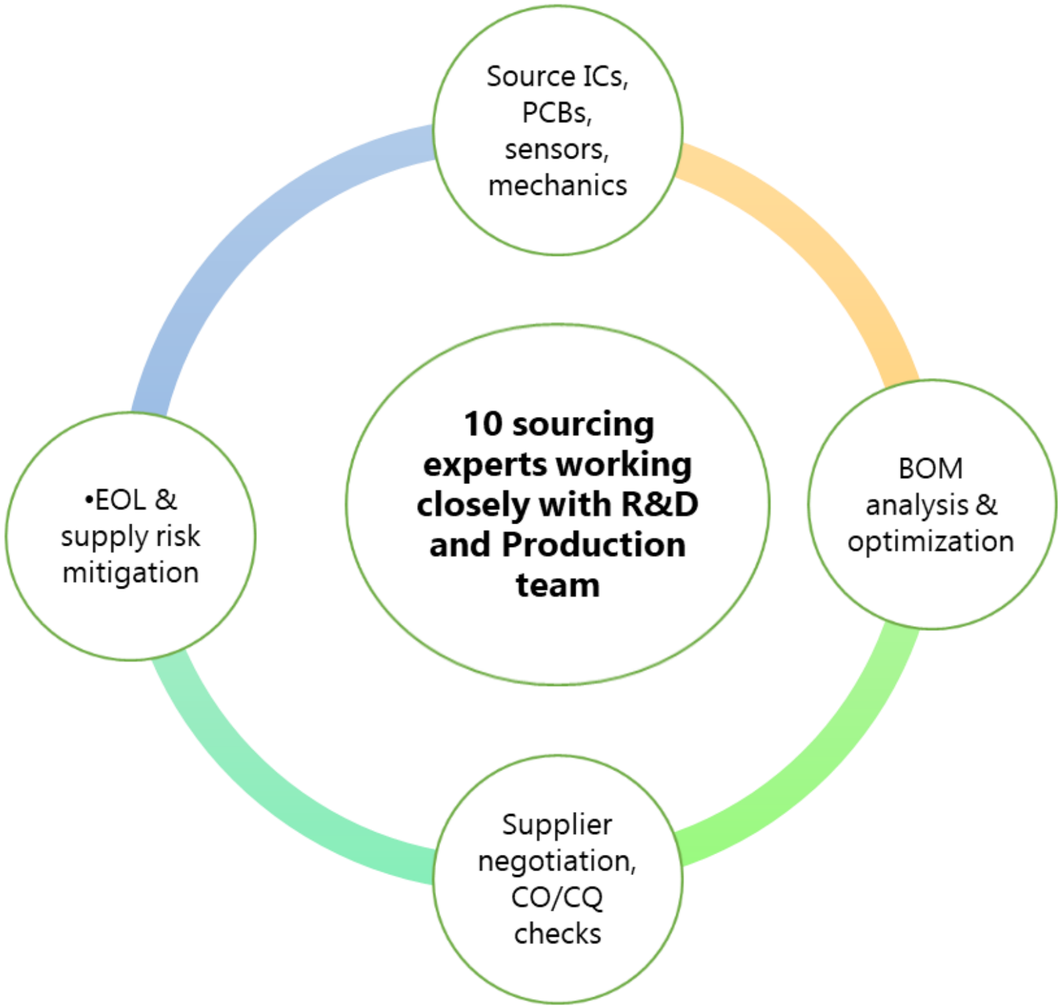
AI that Sees, Thinks, and Acts

Tech stacks

Programming Language	Computer Vision	Deep Learning Frameworks	Data Management	Database	Hardware Accelerator
C++, python	ANPR	TensorFlow, PYTORCH, ONNX, PaddlePaddle		ORACLE, SQL, Postgre SQL, MySQL	NVIDIA, Ambarella™, XILINX

Supply Chain

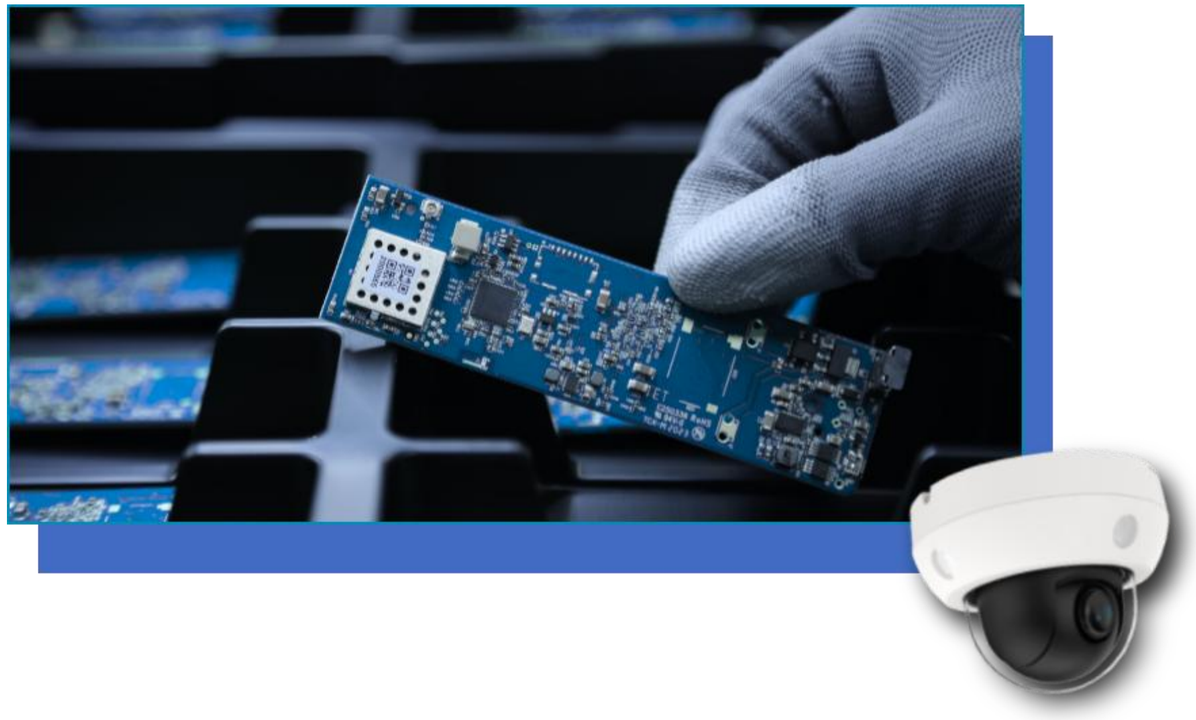
Fueling innovation through strategic sourcing





Supply Chain

Fueling innovation through strategic sourcing



Cost control

- Optimize budget and ensure profit margins in production.
- Focus on reducing unnecessary costs while maintaining product quality.

Risk mitigation

- Implement strategies to minimize risks through supplier assessments and contingency plans.
- Ensure smooth supply process by reducing disruptions and delays.

Sustainability compliance

- Ensure all components meet sustainability standards.
- Full compliance with regulatory standards like NDAA, CE, RoHS, and REACH.

Our partners

An extensive network of both international and domestic suppliers





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Smart Manufacturing Plant

Green. Smart. Scalable.

2 SMT LINES

2 ASSEMBLY LINES

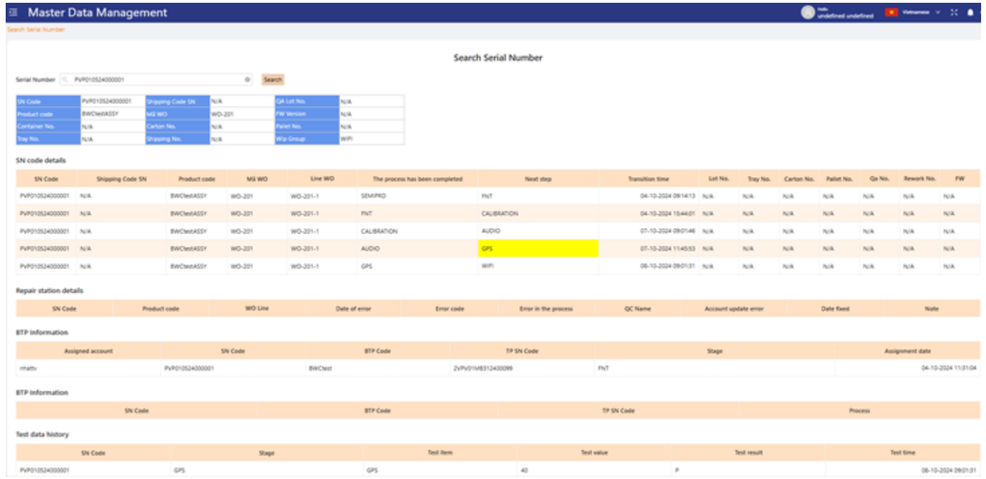
Monthly Capacity: 95,000 cameras

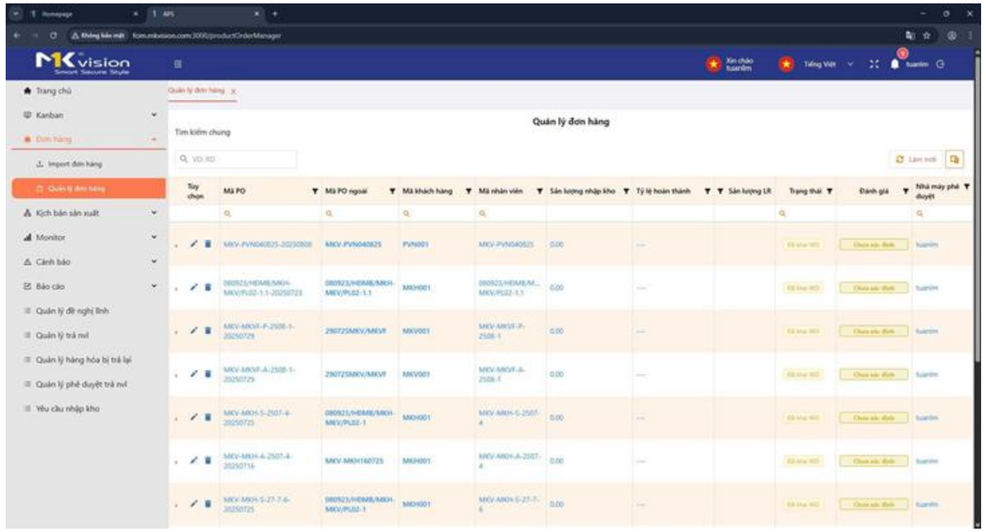
Smart Manufacturing Plant

Green. Smart. Scalable.

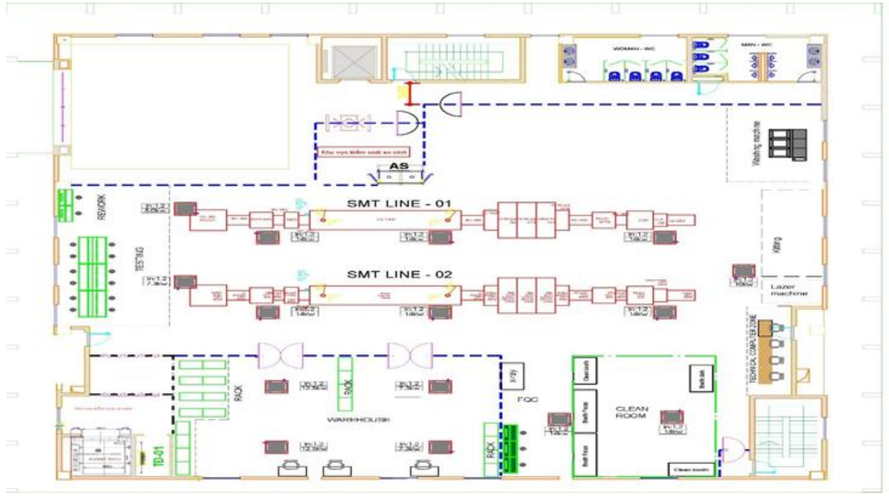
Smart Operations with the FCIM System

- **MDM** - Master Data Management
- **APS** - Advanced Production Scheduling
- **WMS** - Warehouse Management System
- **QMS** - Quality Management System



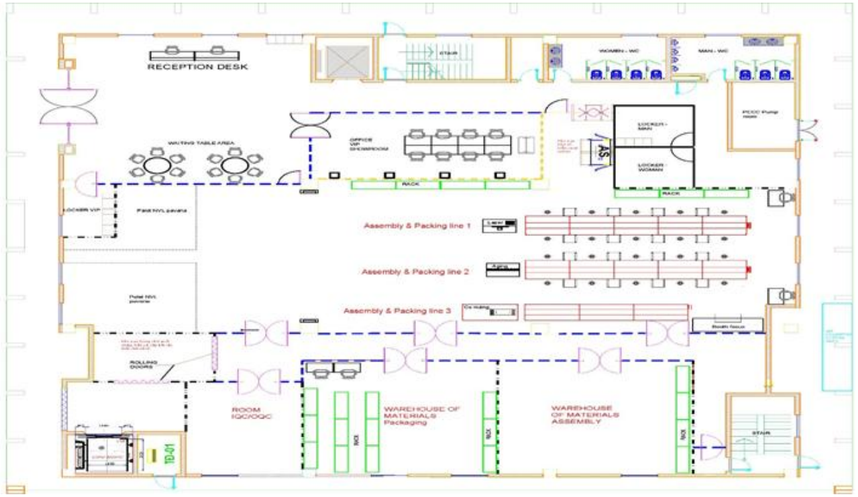


Overall Layout





Floor 1



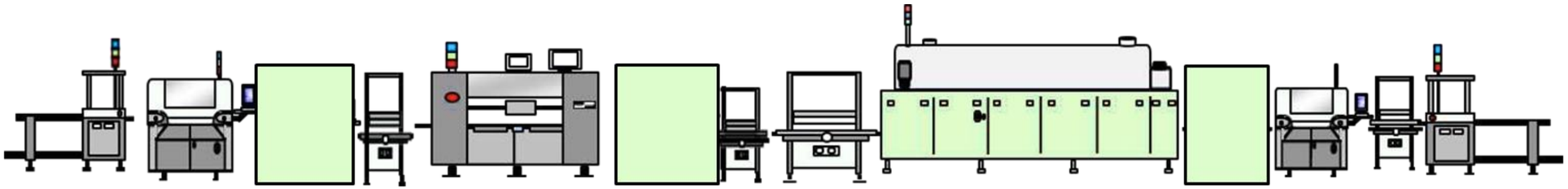


Floor 2

SMT LINE SYSTEM

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SMT line System



INPUT PROCESS



LOADER

PRINTING PROCESS



SOLDER CREAM PRINTING

INSPECTION PROCESS



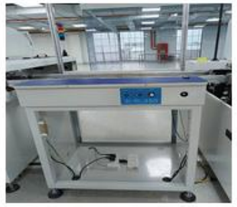
SOLDER CREAM INSPECTION

MOUNTING PROCESS



MOUNTING

INSPECTION PROCESS



MANUAL INSPECTION

REFLOW PROCESS



REFLOW

INSPECTION PROCESS



AOI

OUTPUT PROCESS



UN- LOADER

QC Equipments



X-Ray Inspection



LCR Measurement Device



Pro'skit MA-1209Li 220V Magnifier



Microscope



Electron microscope with screen connection



Caliper



Thichness gauge



Smart tweezers LCR Reader

QC Check

▶ Verification purpose.

- Test the product's durability under different conditions such as: Salt concentration, water pressure, temperature, humidity,...

- Assess whether product quality meets requirements based on quality standards related to IP67, IP68, ISO 9227:2017,....

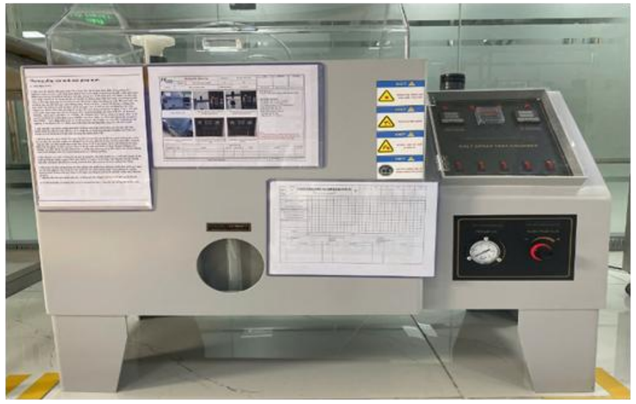




Air leak tester machine



Water-proof machine



Salt spray machine



Environment cabinet machine

Service Offerings

01

Manufacturing only

You provide all components, tools, and processes. We manage assembly and quality control based on your specs.

→ Ideal for clients moving mass production to Vietnam while keeping full supply chain control.

02

Material Sourcing & Manufacturing

You provide BOM, tools, and processes. We source materials, handle production, and ensure QC.

→ Best for clients who need local procurement support in Vietnam.

Service Offerings

03

DFM & Manufacturing

You provide BOM and production specs. We design tools, source parts, manufacture, and manage QC.

→ Perfect for NPI-ready products needing production optimization.

04

Prototyping to Mass Production

You provide complete product designs. We prototype, refine for manufacturability, then scale to mass production.

→ Great for clients needing validation before scaling.

Service Offerings

05

Collaborative Development

You share requirements and assign engineers. We co-develop, handle sourcing, and manage full production.

→ Suited for clients with limited hardware/camera development experience.

06

End-to-End Development & Production

You define the product, target cost, and timeline. We manage everything: R&D, ID, sourcing, tooling, production, and QC.

→ Ideal for clients without internal R&D, seeking a full turnkey solution.

Surveillance solution

Camera lineup for buildings and factories









NDAA COMPLIANT













Product Overview

Advanced Biometric Card Reader – M6



Feature:

- Read contactless cards (NFC)
- Operating on the Linux platform, Ambarella CV25 Quad-core 1.0 GHz processor
- Integrated biometric module (face, fingerprint) to perform real face/fingerprint comparison with face/fingerprint stored in the chip
- 5 inchs touch screen
- Support Wifi, Ethernet

Application:

- Access control at buildings, hotels, public areas...
- Identify and verify customers when making transactions at the counter

Product Overview

Body Worn Camera





Feature:

- 1080 Full HD video recording with AI processing chip (NPU) for face recognition
- NFC-based ID verification (e-ID, e-passport)
- 4G LTE, GPS & GLONASS/A-GPS connectivity
- IP65 dust and water resistance
- 12+ hours operation per charge
- Compact 205g design for mobility

Application:

- Access control at buildings, hotels, public areas...
- Identify and verify customers when making transactions at the counter

Product Overview

V-Surveillance serial

Feature:

- 5MP camera with IP67 protection and ONVIF/RTSP support
- ZoneIQ – Intrustion detection and boundary – crossing alerts
- Lenslock – Camera tampering detection (scene or color change alerts)
- Open API for seamless software intergration
- Continuous recording with secure storage, Wi-Fi/Ethernet support



Application:

- Area surveillance and intrusion monitoring in residential, commercial, and industrial zones
- Reliable video evidence storage for investigation and security management
- Intergration with ONVIF – based or custom management software for centralized control

Product Overview

Video conference

01 | Outstanding video-audio quality

02 | Connect instantly

03 | Compatible with most devices

04 | Excellent room coverage (C360)

05 | Speaker tracking (C360)

06 | Privacy protection (C100)

SC360 – Room



SC100 – Small Group



W100 – Personal





Thank you all for being here today!

MK Vision Team

Ha Noi, 2025